

Explainable AI in Financial Supervision

*Research Information Deck for
Banca d'Italia*

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About us

MSCA Industrial Doctoral Network on Digital Finance



This EU-funded Industrial Doctoral Network, running from 1 January 2024 to 31 December 2027, supports 17 fully funded PhD positions focused on the intersection of finance, data science, and artificial intelligence. Funded under the Marie Skłodowska-Curie Actions (MSCA), the network brings together academic institutions, financial companies, technology providers, and policy organizations to jointly train the next generation of researchers in digital finance.

The project on Explainable AI (XAI) is hosted by **University of Naples Federico II** under the supervision of [Dr. Maria Iannario](#) and [Dr. Ekaterina Svetlova](#) and [Dr. Doina Bucur](#) from **University of Twente, Netherlands**.



[Project link](#)

Our research team

01



Patricia Marcella Evite

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BS Economics, University of the
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02



Maria Iannario

***Professor of Statistics, UNINA**
Principal Investigator of IRP9 in
MSCA DIGITAL*

03



Ekaterina Svetlova

***Professor of Accounting and Finance,**
Faculty of Behavioural, Management
and Social Sciences, University of
Twente, Netherlands*

04



Doina Bucur

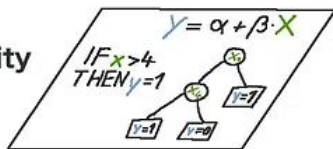
***Assistant Professor in network data
science, University of Twente,
Netherlands***

Humans



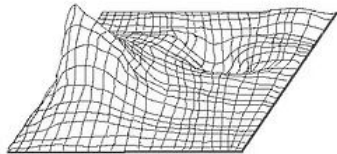
↑ inform

Interpretability
Methods



↑ extract

Black Box
Model



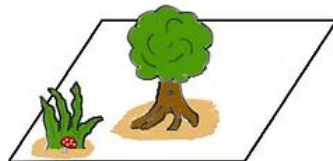
↑ learn

Data

K	X	K	.	.	.	.	X
13	2	0	.	.	.	.	13.00
15	4	0	.	.	.	.	15.00
7	1	0	.	.	.	.	7.00

↑ capture

World



Explainable AI (XAI) tries to make surrogate explanations for complex machine learning models, but they lead to incomprehensible formats for non-technical audience.

The problem

Finance increasingly relies on advanced machine learning models that deliver strong performance but often function as “black boxes.” Explainable AI (XAI) aims to make these systems more transparent and understandable. **However, these explanations are not universally useful.**

Regulators, compliance teams, business users, developers, and customers need different levels of detail and different kinds of evidence. The key challenge is not only explaining AI, but making explanations fit the audience and accountability context as suggested by previous authors ([Hadji-Misheva et al., 2021](#); [Maxwell & Dumas, 2023](#))

The SHAP plot below is an example of an XAI model.

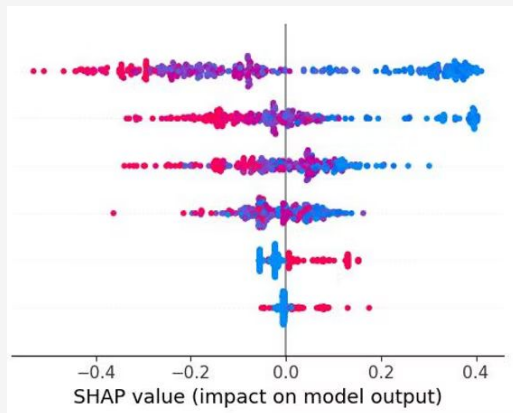


Image credit: [Molnar \(2025\)](#)

But according to our preliminary findings, they are not intuitive to bankers:

"The business would not understand it (blackbox models). They were so used to the regression coefficients."

AI Product Manager, Asia

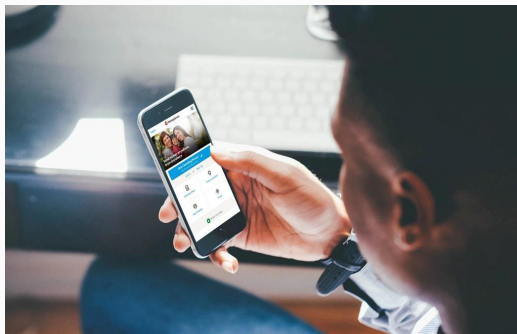
Our mission: find out what is a good explanation for you?

The project moves beyond evaluating existing AI tools to **empirically profile the preferences of the people who actually use them**. We aim to map the requirements of explainable systems across **three key stakeholders**:



01

Finance Professionals: Who need explanations to make and defend decisions



02

Public: Who need friendly explanations to calibrate trust in automated services and empower them to contest unfair



03

Regulators: Who require explanations contextualized with legal goals and compliance and us AI for risk-based supervision

YOU are part of our third target audience.

Data request

Why participate?

Little is known about regulator preference for AI explanations. **Help us change that.**

Shape practical guidance

Ensure supervisory needs inform emerging explainability frameworks.

Benchmark and learn

Compare approaches across central banks (EU + extra) and identify common pain points and good practices.



Support governance readiness

Help translate explainability into model governance, validation expectations, and supervisory communication.

Who can participate?

Generally,
participants who
either
✓ use AI in
supervisory or
forecasting work,
and/or
✓ set expectations
for banks'
AI/automated
decisions

AI use case / task

Typical AI methods / tools

Predictive analytics & risk forecasting

Supervised Machine Learning (ML); time-series models; deep learning (where used)

Early warning & anomaly detection

Unsupervised learning; outlier detection; clustering

Document processing & summarisation

Natural Language Processing (NLPs); Large Language Models (LLMs like ChatGPT); Optical character recognition (OCR)+ information extraction

Market surveillance & monitoring

Anomaly detection; pattern recognition; rules + ML hybrids

Comparing assessments & narratives

NLP similarity; topic modelling; embeddings / clustering

Interconnectedness & systemic risk mapping

Graph / network analytics

Example of roles



Prudential supervision

Uses supervisory data/tools for risk assessment, SREP inputs, onsite/offsite monitoring

Policy or AI governance specialists

Develops or interprets supervisory expectations, guidance, or AI governance requirements

Data / supotech / analytics professionals

Builds or uses supervisory analytics/AI tools (e.g., dashboards, anomaly detection, NLP)

Model developer, validator

Reviews model documentation, validation reports, monitoring, limitations, and controls

We invite you to participate in two ways:

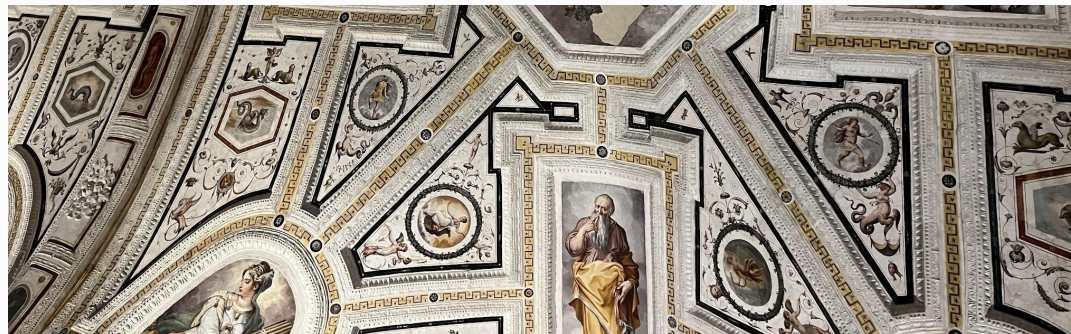


01 Answer anonymous **survey** internally
Link open until **March 15, 2026**
www.tinyurl.com/xai-finpro

02 Nominate 2-3 candidates for expert **interviews**

45-60 minutes individually, semi-structured guide
Between March - April 2026, at your convenience

Ethical considerations



01

Recording and attribution

- Audio will only be stored until transcript is validated; no personal identifiers and all data are confidential
- Informed consent with preferences on attribution will be provided

02

Voluntary participation

May skip or withdraw questions anytime

03

Use of collected info

Solely for academic use
All results in open-access journals and findings reported in aggregated and anonymized formed

04

Adherence to GDPR and UNINA ethics

Reviewed and approved by the Ethics Committee for Research Involving Human Subjects in Biomedical Fields (CERSUB) of the University of Naples Federico II, on September 25, 2025, with reference number PG/2025/0133225.

Join a **global** study.

Here is a list of confirmed participating regulatory institutions in our survey & interviews.



European
Central
Bank



Mongolbank
Mongolia



Bangko
Sentral ng
Pilipinas
Philippines



Central
Bank of the
Republic of
Uzbekistan



Central
Bank of
Kenya



Qatar
Financial
Center
Regulatory
Authority

Next steps &
timeline

I'm in. What's next?



01

Fill out the **survey** here. The link open until **March 15, 2026**.

www.tinyurl.com/xai-finpro



02

If you are interested in the **interviews**, please reach out by email: patriciamarcella.evite@unina.it
Or alternatively, you can book an *online* interview slot directly here:

<https://calendar.app.google/3ovv5qzmC3tWZ7Gp9>

For in-person interviews in Rome, please reach out by email. You can view the informed consent [here](#) and list of potential questions [here](#).

Next steps

March 15

Survey closes



April 30

End of interviews



May-June

Validation of transcripts



July onwards

Analysis and submission of publications.
Participants will be given pre-prints of outputs in advance.

Grazie mille.



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